Is Liver Transplantation Appropriate for Patients with Potentially Resectable De Novo Hilar Cholangiocarcinoma?

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BACKGROUND: Liver transplantation (LTX) is curative for selected patients with hilar cholangiocarcinoma (HC) in the setting of sclerosing cholangitis. However, the outcome of LTX vs liver resection (RTX) for patients with de novo HC remains unclear.

STUDY DESIGN: Patients with de novo HC treated by protocol LTX (n = 90) or RTX (n = 124) between 1993 and 2013 were reviewed. Based on preoperative imaging, RTX was pursued for Bismuth type III HC and LTX for unresectable Bismuth type IV.

RESULTS: Unadjusted analysis showed that overall survival after operation was greater for LTX than RTX (p = 0.003). One-, 3-, and 5-year overall survival rates, respectively, were 90%, 71%, and 59% for LTX and 81%, 53%, and 36% for RTX. Survival was not different between LTX and RTX after adjusting for patient age, lymph node metastases, and tumor size. After postoperative pathologic review, HC after RTX was reclassified as Bismuth-Corlette (B-C) IV, based on the necessity of multiple biliary anastomoses in 40 patients to more accurately compare treatment outcomes. Overall survival was greater after LTX than RTX (p = 0.039) for patients with Bismuth-Corlette IV HC.

CONCLUSIONS: Patients with clearly resectable de novo HC should be treated with resection because there is no evidence that they would fare better with LTX. Patients with locally unresectable de novo HC, meeting criteria for our protocol, should be treated with LTX. The decision to proceed with RTX or LTX for patients with borderline resectable de novo HC remains difficult, but our results suggest that patients with B-C type IV HC might be best treated with transplantation, if they are excellent transplant candidates. (J Am Coll Surg 2015;221:130−139. © 2015 by the American College of Surgeons)

Hilar cholangiocarcinoma (HC) remains a formidable surgical challenge because of its propensity for invasion of adjacent liver parenchyma, encasement of vasculature, and metastases to regional lymph nodes. Locally advanced disease at diagnosis and surgical inaccessibility has resulted in low resectability rates and poor patient survival. Despite these limitations, liver resection (RTX) and liver transplantation (LTX) provide the only possibilities for cure.

The extent of ductal involvement, hepatic parenchymal invasion, and portal venous and hepatic arterial encasement often preclude resection. The Bismuth-Corlette system (B-C) provides a preoperative estimate of ductal extension and is often used to determine the feasibility of resection. Historically, lesions classified as B-C types IIIa and IIIb were considered resectable by resection of the extrahepatic bile duct and right or left hepatectomy; cancers extending into bilobar segmental intrahepatic ducts (type IV) were considered unresectable.[1] Standard resections for HC have been associated with a 5-year survival of 10% to 40%.[2-4] Advances in surgical technique, including portal venous and hepatic arterial reconstruction, have enabled a larger number of patients to undergo resection.[5] Nevertheless, many patients present with locally advanced HC precluding even aggressive RTX, and LTX is their only option.

Disclosure Information: Nothing to disclose.

Presented at the Western Surgical Association 122nd Scientific Session, Indian Wells, CA, November 2014.

Received December 17, 2014; Revised January 28, 2015; Accepted January 28, 2015.

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Published by Elsevier Inc.

http://dx.doi.org/10.1016/j.jamcollsurg.2015.01.064
ISSN 1072-7515/15

Abbreviations and Acronyms
B-C = Bismuth-Corlette system
HC = hilar cholangiocarcinoma
HR = hazard ratio
LTX = liver transplantation
PSC = primary sclerosing cholangitis
RTX = liver resection

We and others have used the strategy of neoadjuvant therapy with high dose radiotherapy, chemosensitization, and operative staging followed by LTX for selected patients with unresectable HC.[6-10] Although excellent results have been demonstrated with neoadjuvant therapy and LTX,[6-11] there is a significant difference in survival between patients with HC arising in the setting of primary sclerosing cholangitis (PSC) and those with HC arising de novo. Resection is rarely possible for patients with HC arising in the setting of PSC. These patients often have underlying liver disease that precludes resection.[12,13] They also have a propensity for multicentric carcinoma. Neoadjuvant therapy and LTX is the only treatment that can effectively treat both the cancer and the underlying liver disease and is clearly the treatment of choice for these patients.

The choice of therapy for patients with HC arising de novo is controversial and unclear. Selection of appropriate surgical therapy for patients with de novo HC requires careful comparison of the results of resection and transplantation. Other factors to consider include resource use, donor organ availability, and morbidity. We wondered whether we should expand our current selection criteria for LTX to include patients with potentially resectable or borderline resectable HC. We reviewed our recent experiences with resection and transplantation for patients with de novo HC with the specific aim of determining which patients, if any, with potentially resectable or borderline resectable HC might be better treated with neoadjuvant therapy and LTX. Our intention was to better define operative selection criteria.

METHODS

This study was performed with approval of the Mayo Clinic Rochester Institutional Review Board. Data were acquired from patient medical records, outside medical records, and from a database maintained on all patients enrolled in our HC transplant protocol.

We reviewed data for all patients with HC treated by RTX or the neoadjuvant therapy and LTX protocol at Mayo Clinic Rochester between January 1, 1993 and July 1, 2013. An experienced hepatobiliary surgeon evaluated all patients with de novo HC to assess resectability before enrollment in the LTX protocol. Selection criteria for the LTX protocol have been described previously and require that patients have locally unresectable HC without evidence of either nodal or distant metastatic disease, no previous treatment, and no transperitoneal biopsy or needle aspiration of HC.[9,14,15] Patients must be suitable transplant candidates. Patients who met initial eligibility criteria underwent protocol neoadjuvant therapy, which has been described in detail in previous publications.[9] All patients completing neoadjuvant therapy and remaining within selection criteria underwent a staging operation with biopsy of regional (pericholedochal and proper hepatic artery) lymph nodes near the time of LTX. Regional lymph node involvement, peritoneal metastases, intrahepatic metastases, or local invasion of adjacent organs or tissues precluded LTX.

The standard treatment of patients with resectable de novo HC included hemihepatectomy with or without anatomic (segments 1, 4) or nonanatomic contralobar extension, hepatoduodenal ligament lymphadenectomy and bile duct resection and Roux-en-Y hepaticojejunostomy. Segmental or partial portal venous resection with reconstruction was performed as necessary to achieve R0 resection. Patients undergoing resection were classified as B-C type IIIA or IIIB if the ductal resection resulted in a negative margin to a single intrahepatic duct.[1] In these patients, a single duct was anastomosed to the Roux-en-Y jejunal limb. In patients who required resection of multiple segmental or sectoral ducts in the remnant to obtain negative ductal margins, B-C type was reclassified postoperatively as B-C type IV to more accurately define ductal extension for comparison of outcomes between resection and transplantation. In patients in whom transected intrahepatic ductal apertures were closely approximated, adjacent duct walls were anastomosed together to provide a common aperture to facilitate a single, bilio-enteric anastomosis. If the transected intrahepatic ducts could not be approximated, we performed separate bilioenteric anastomoses.

All resection patients had histologic confirmation of biliary adenocarcinoma on resected specimens. No patient had distant lymphatic or other metastatic disease at the time of resection. Postoperative pathologic staging could not be performed completely because of the effects of neoadjuvant therapy in the LTX group. Despite the presence of a TNM system specific for HC, all patients undergoing LTX presumably had stage IVA TNM stage disease based on the T4 stage definition (BC type IV) of the American Joint Committee on Cancer, 7th Edition.

Progression-free survival was calculated as the time between RTX or LTX and evidence of recurrent or

metastatic cholangiocarcinoma or patient death from any cause, whichever occurred first. Overall survival was calculated as the time between RTX or LTX and patient death. The time from entry into the transplantation protocol (start of neoadjuvant therapy) until patient death was used for intention-to-treat analyses.

All statistical analyses were performed using STATA 12 (Stata Corp). Differences between groups were analyzed using the unpaired *t*-test for continuous variables and by the chi-square test or continuity correction method for categorical variables. Survival curves for patient survival were generated using the Kaplan-Meier method and compared by the log-rank test. Prognostic variables were identified by a univariate Cox proportional hazard model. In addition, we performed a multivariate Cox proportional hazard model with backward stepwise selection to predict patient survival. A liberal retention criterion of $p < 0.10$ level of significance during the backward stepwise search was used.[16] A propensity score analysis was performed and included age, sex, tumor size and grade, lymph node involvement, and portal vein. All statistical tests were 2-sided, and differences were considered significant when $p < 0.05$.

RESULTS

Ninety-nine patients underwent RTX and 54 patients underwent LTX for de novo HC. Patient demographics for the 2 groups are shown in Table 1. Mean age was greater in the RTX group (63.3 ± 11.1 years) compared with patients in the LTX group (54.1 ± 9.4 years) ($p < 0.001$). The predominance of male patients was similar between the RTX and LTX groups (65% and 74%, respectively, $p = 0.23$). Patients in the RTX group had a greater BMI than patients in the LTX group: 26.9 ± 5.4 kg/m^2 and 24.8 ± 4.9 kg/m,^{2} respectively ($p = 0.02$).

Left hepatectomy with caudate lobectomy was performed in 40 patients and left hepatectomy without caudate lobectomy was performed in 4 patients. Right hepatectomy with caudate lobectomy was performed in 8 patients, right hepatectomy without caudate lobectomy was performed in 42 patients, right trisectorectomy in 1 patients, and a mesohepatectomy in 1 patient. A single bilio-enteric anastomosis to a single lobar duct from the remnant was performed in 59 patients. Bilio-enteric reconstruction for multiple ducts in the remnant was performed in 40 patients, including 3 hepaticojejunal anastomoses in 3 patients, 2 hepaticojejunal anastomoses in 16 patients, and a single hepaticojejunal anastomosis to a single ductal orifice after approximation of 2 or more transected segmental or sectoral ducts from the remnant in 21 patients.

Fifty-four patients underwent LTX, 36 (67%) with a deceased donor liver and 18 (33%) with a living donor liver (17 right and 1 left grafts). Two patients underwent concurrent pancreaticoduodenectomy with LTX because of a positive margin in the distal bile duct. One patient ultimately developed recurrence in the retroperitoneum at 16 months after transplant and died; the other patient is alive and recurrence free.

The frequency of R0 resection in the LTX group (100%) was greater than that of the RTX group (92%) ($p = 0.02$). Mean pathologic tumor diameter was greater in the RTX group (2.8 ± 1.5 cm) than in the LTX group (1.0 ± 1.8 cm) ($p < 0.001$). Actual size of the HC in the LTX group was likely underestimated by explant pathologic findings because of the effect of neoadjuvant therapy. Residual cancer in the LTX group was present in 37 of 54 (69%) patients. As would be expected by LTX

Table 1. Patient Demographics and Pathologic Characteristics in Resection and Liver Transplantation Groups

Patient demographics	Resection (n = 99)	Transplantation (n = 54)	p Value
Age, y, mean ± SD	63.3 ± 11.1	54.1 ± 9.4	<0.001
Sex, male, n (%)	64 (65)	40 (74)	0.23
BMI, kg/m^2, mean ± SD	26.9 ± 5.4	24.8 ± 4.9	0.02
Margin negative, R0, n (%)	90 (92)	54 (100)	0.05
Tumor diameter, cm, mean ± SD	2.8 ± 1.5	1.0 ± 1.8	<0.001
Positive nodes, N1, n (%)	37 (38)	1 (2)	<0.001
Tumor grade, n (%)			
Well, G1	3 (3)	1 (2)	0.66
Moderate, G2	20 (20)	16 (30)	0.19
Poor, G3	63 (63)	19 (35)	<0.001
Undifferentiated, G4	12 (12)	2 (4)	0.08
Indeterminate/not viable	1 (1)	16 (30)	<0.001

BMI, body mass index.

protocol design, lymph node metastases were significantly more frequent in patients in the RTX group than in the LTX group (38% and 2%, respectively, $p < 0.001$). Hilar cholangiocarcinoma with poorly differentiated histopathology was more frequent among patients in the RTX group than the LTX group ($p < 0.001$), and 16 (30%) of the explanted livers in the LTX group had indeterminate cancer grades or no detectable residual cancer as a result of neoadjuvant therapy.

Median follow-up was 26 months in the RTX group and 43 months in the LTX group. Patients undergoing LTX had a greater overall survival than patients undergoing RTX ($p = 0.003$; Fig. 1). One-, 3-, and 5-year overall survival rates were, respectively, 90% (95% CI 79% to 95%), 71% (95% CI 55% to 82%), and 59% (95% CI 44% to 72%) in the LTX group and 81% (95% CI 74% to 89%), 53% (95% CI 41% to 64%), and 36% (95% CI 24% to 47%) in the resection group. Progression-free survival was also greater in the LTX group than in the RTX ($p < 0.001$; Fig. 2). One-, 3-, and 5-year progression-free survival rates were 87% (95% CI 74% to 95%), 64% (95% CI 48% to 76%), 54% (95% CI 39% to 66%) in the LTX group and 74% (95% CI 63% to 82%), 48% (95% CI 37% to 59%), and 29% (95% CI 18% to 40%) in the resection group, respectively.

A Cox univariate regression analysis for overall survival revealed the following factors associated with a decrease in overall survival: resection (with transplantation as reference) hazard ratio (HR) 2.2 ($p = 0.001$); age (per 10-year increase), HR 1.48 ($p < 0.001$); tumor diameter, HR 1.17 ($p < 0.001$); positive nodal status, HR 2.1 ($p = 0.002$); and tumor grade, HR 2.4 ($p < 0.001$). A multivariate Cox proportional hazard model with backward stepwise selection revealed 2 factors associated with a decrease in overall survival: age, HR 1.39 ($p = 0.003$) and tumor grade, HR 1.85 ($p = 0.02$) (Table 2). An additional analysis using propensity score consisting of the aforementioned variables was used comparing survival in the RTX and LTX groups; mean propensity score was different between the treatment groups before matching: 0.21 ± 0.25, respectively, ($p = 0.001$). After matching on propensity score, the overall survival was not significantly different by operation (HR 1.8, 95% CI 0.79 to 4.10, $p = 0.16$).

A subgroup analysis comparing 57 patients who underwent RTX with 53 patients who underwent LTX and had no lymph node metastases and negative margins of resection, demonstrated no significant difference in survival between the RTX and LTX groups ($p = 0.41$, Fig. 3). To further compare HC between groups and more clearly define variants of B-C type III HC, the B-C type was reclassified as B-C type IV in 40 patients in the resection group when intraoperative resection of sectoral or segmental ducts in the remnant was required to obtain negative margins. Patient survival was significantly greater for patients with B-C type IV HC in the LTX group compared with those in the RTX group ($p = 0.039$, Fig. 4).

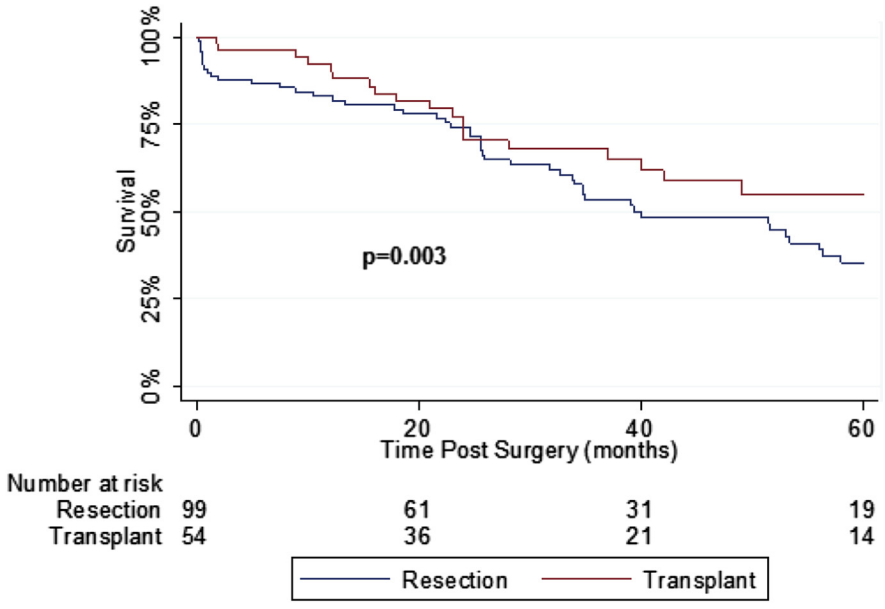


Figure 1. Kaplan-Meier survival estimates. Overall survival in patients undergoing resection vs liver transplantation.

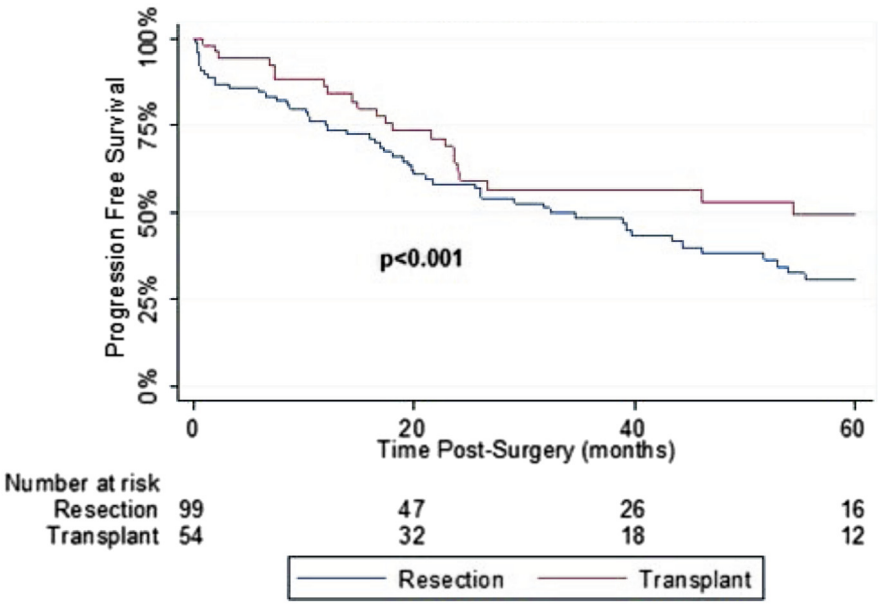


Figure 2. Kaplan-Meier survival estimates. Progression-free survival in patients undergoing resection vs liver transplantation.

Finally, an intention-to-treat analysis was performed on all patients with de novo HC who were explored for resection (n = 142) or enrolled in the LTX protocol (n = 90) during the study period. The HC was either confirmed pathologically or met criteria for diagnosis of HC in the LTX protocol. Forty-three patients (30%) who underwent operation for intended resection were found unresectable. Reasons for unresectability included: extent of vascular encasement (14 patients, 33%), peritoneal metastases (12 patients, 30%), distant lymph node metastases (10 patients, 23%), and intrahepatic metastases (7 patients, 16%). Of the 90 patients who were enrolled in the LTX protocol, 7 patients (8%) developed tumor progression before operative staging and 2 patients (2%) were excluded from staging because of recurrent sepsis and deconditioning. Of the 79 patients (88%) who completed neoadjuvant therapy and underwent operative staging, 25 patients (28%) had operative findings (13 had peritoneal metastases with or without lymph node involvement and 12 had lymph node involvement) that precluded liver transplantation. In total, 54 patients (60%) who entered the protocol ultimately underwent transplantation. Intention-to-treat analysis showed that overall survival was greater for patients in the LTX protocol than those who underwent operation intended to be RTX (p = 0.007, Fig. 5).

Table 2. Univariate and Multivariate Backward Step-Wise Cox Regressions

	Univariate		Multivariate	
Variables	Hazard ratio	p Value	Hazard ratio	p Value
Age, per 10-y increase	1.48	<0.001	1.39	0.003
BMI, kg/m^2	1.06	0.13		
Sex, male	1.28	0.3		
Resection/transplantation	2.20	0.001		
Margin negative, R0	1.45	0.43		
Tumor diameter, cm	1.17	<0.001		
Portal vein invasion	1.25	0.5		
Positive nodes, N1	2.10	0.002		
Tumor grade, G3	2.40	<0.001	1.85	0.02

BMI, body mass index.

DISCUSSION

The results of our study affirm the efficacy of both RTX and LTX for treatment of patients with de novo hilar cholangiocarcinoma. Our key findings were: there is a difference in overall and progression-free survival among patients undergoing neoadjuvant therapy and liver transplantation compared to patients undergoing resection by both post-surgery survival as well as that by an intention-to-treat analysis; and on subgroup analysis, survival was less among patients who were reclassified as having B-C type IV HC ("borderline resectable") after RTX than after LTX. Survival, however, did not differ between treatment groups among patients with node-negative HC and margin-negative approaches. Other findings included affirmation of previously identified factors associated with poor survival: older age, larger tumor size, regional

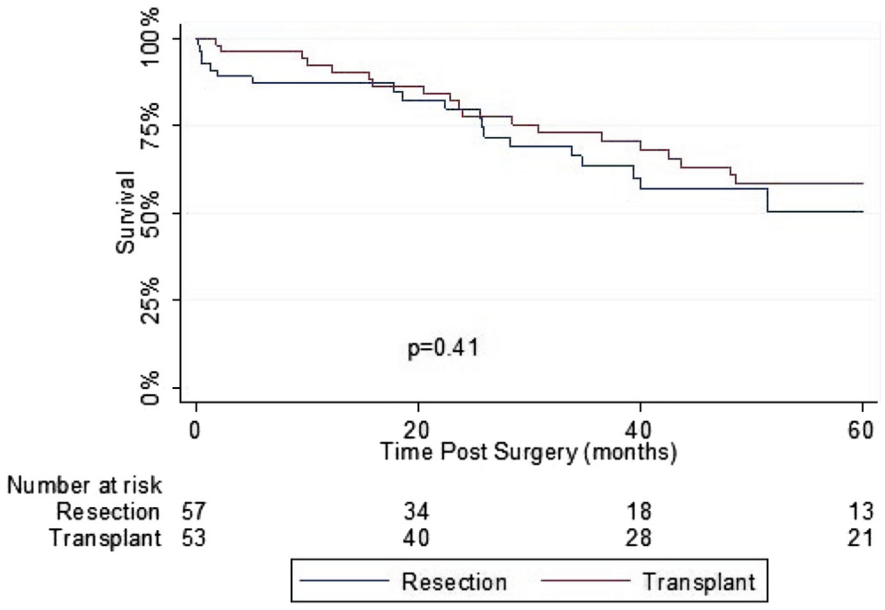


Figure 3. Kaplan-Meier survival estimates. Overall survival in patients undergoing resection and liver transplantation with N0, R0 disease.

lymph node involvement, poorly differentiated histology by univariate analysis and larger tumor size, and poorly differentiated histology by multivariate analysis.

Hilar cholangiocarcinoma is one of the most difficult management problems faced by surgeons. Complete removal of tumor is necessary to achieve a cure and prolong patient survival. The surgical choice of operation—RTX or LTX—depends on whether the cancer is resectable, that is, R0 removal is expected preoperatively. Unfortunately, lymphatic metastases, hepatic parenchymal invasion, and vascular encasement frequently preclude resection, and the presence of 1 or more of these conditions is difficult to assess before operation. Selection of operative approach is based primarily

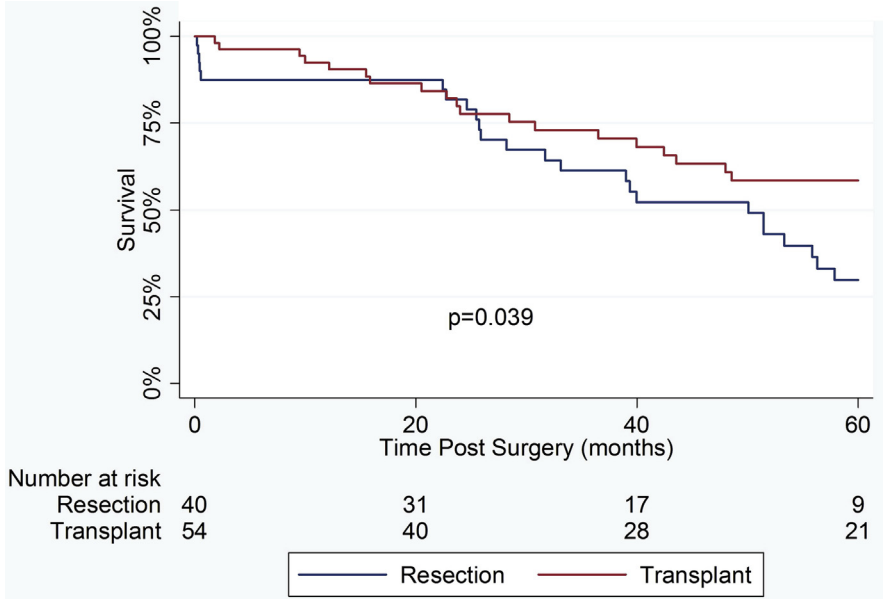


Figure 4. Kaplan-Meier survival estimates. Comparison of Bismuth IV patients undergoing resection vs liver transplantation.

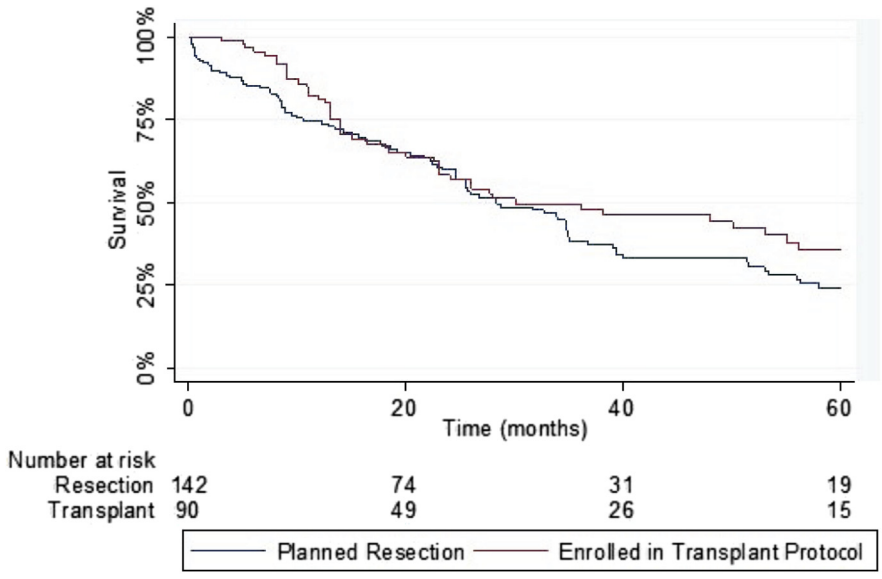


Figure 5. Kaplan-Meier survival estimates. Intention-to-treat overall survival in all patients taken to the operating room with planned resection vs those enrolled in liver transplantation protocol, $p = 0.007$.

on cholangiography, cross-sectional imaging, and clinical performance status. Fit patients with B-C type III HC, with or without nodal metastases and limited vascular involvement, are candidates for RTX, and fit patients with B-C type IV HC without nodal involvement regardless of vascular involvement are candidates for LTX. Admittedly, resection has been used for some patients classified as B-C type IV HC, which continues to confound the definition of unresectability for HC.[17,18]

The decision whether to proceed with RTX or LTX is critical. The decision must be made at the outset of therapy because initiation of either approach excludes the other. Patients explored for RTX and found to have unresectable disease have not done well with subsequent LTX. Currently, these patients are excluded from LTX by our selection criteria. Moreover, a large number of patients selected for RTX have positive ductal margins after RTX and have a noncurative R1 resection. If these patients had no lymph node metastases, they would have been candidates for LTX had RTX not been undertaken. In contrast, patients undergoing neoadjuvant therapy as per transplant protocol are not candidates for RTX due to the local-regional extent of perihilar and hepatic inflammation and damage. The radiation given in the transplant arm is significant. At the time of LTX, the whole radiation field is fried, so once patients have received the transplant neoadjuvant protocol, proceeding to RTX is really not an option. No patients have been "beyond" transplant inclusion criteria and then been downstaged into the protocol with neoadjuvant therapy. The strict criteria of our LTX protocol were developed because patients beyond these criteria have poor outcomes. Exception points, in terms of transplant listing, have only been allotted for patients meeting the previously mentioned criteria. This is different than the case with LTX for hepatocellular carcinoma, where certain downstaging protocols are allowed.

An intention-to-treat analysis is essential for comparing the efficacy of RTX with that of neoadjuvant therapy and LTX. Our intention-to-treat analysis did demonstrate a difference in survival between patients with de novo HC entered into the LTX protocol or explored for RTX. In part, the difference in intention-to-treat survival is likely due to the fact that early survival for the transplant group reflects wait time for donor availability before LTX in contrast to patients immediately undergoing RTX. However, greater progression-free survival for the LTX group does contribute to late differences in the intention-to-treat survival curve. This difference in survival attests to the efficacy of our selection process. We achieved a 92% R0 resection rate compared with the range of 60% to 80% reported in the literature.[19,20] We also achieved 36% 5-year survival, which compares favorably to the 10% to 40% range reported in the literature.[19-22]

Our results offer no evidence to suggest that patients with clearly resectable de novo HC would be better treated by neoadjuvant therapy and LTX. One could argue that the patients explored for RTX had less advanced disease than those enrolled in the transplant protocol based on the current American Joint Committee

on Cancer 7th Edition staging system and likely would have fared better with transplantation than those patients who underwent transplantation. The merit of this argument can only be resolved by a randomized prospective trial. Comparison of survival of our patients after RTX and LTX, however, suggest that the likelihood of a significant difference would be low because we observed no difference in survival between RTX and LTX for patients with node-negative disease and R0 resection. Importantly, this finding stresses the importance of accurately selecting patients for operation. If node-negative and margin-negative patients could be clearly identified preoperatively, RTX would be preferred.

We acknowledge that there were clear differences in clinical and pathologic findings between our treatment groups. Indeed, these differences were used to select the surgical approach. Our study shows that overall survival was greater after LTX than RTX, but the patients receiving transplants harbored smaller HC and infrequent (by selection) nodal metastases. The impact of neoadjuvant therapy on HC size cannot be assessed. Nevertheless, critical examination of our experience proved to be helpful in identifying which, if any, of our patients with potentially resectable or borderline resectable HC should be considered for neoadjuvant therapy and LTX instead of RTX. Patients undergoing RTX in this study were all thought to have B-C type III tumors based on preoperative imaging. Forty patients were reclassified as B-C type IV ("borderline resectable") based on postoperative pathologic review that demonstrated tumor extension up both the left and right hepatic ducts, necessitating multiple biliary anastomoses for reconstruction. The B-C type IV tumors, which extend into the secondary biliary radicals on both sides of the liver, have historically been considered unresectable.[1] Studies looking at survival in type IV patients undergoing RTX have demonstrated a median survival of 25 to 29 months; those looking at type IIIA/IIIB have demonstrated a median survival of 27 to 50 months.[2,17,18,23,24] Recent publications have suggested that in select cases, strategies of high hilar resection at the level of secondary or tertiary biliary radicals and reconstruction using various techniques including multiple bilioenteric anastomoses and portal parenchymal-enterostomy, can facilitate resection in these patients.[17-18] There are also recent data suggesting that resection for patients with B-C type IV tumors has significantly worse survival than for patients with B-C type III tumors.[25] The inferior survival was attributed to the greater rate of positive surgical margins (37.1%) in the type IV compared with the type III group (19.5%). In patients with N0 disease who underwent R0 resection, there was no difference in survival between patients with B-C type III and IV disease. In this study, there was no significant difference in survival between patients who underwent RTX and LTX when the groups were restricted to patients with negative margins and negative nodes. However, inferior survival was demonstrated when patients reclassified as B-C type IV were compared with patients who underwent LTX. Therefore, patients with borderline resectable disease might be better treated with LTX than RTX. We believe the concept of "borderline" resectable HC, similar to that for pancreatic ductal cancer, should be further assessed for future treatment strategies.

Imaging has evolved significantly over the time period of the study. This accounts in part for the 40 patients who were preoperatively thought to be B-C type III but were reclassified as B-C type IV after RTX. Previously, our group had used percutaneous transhepatic cholangiography and angiography when assessing resectability. Now we use contrast-enhanced MRI for assessment in all patients. Endoscopy is used to delineate the extent of distal bile duct involvement. Unfortunately, access to the right and left hepatic ducts, to determine proximal extent of the tumor, is limited secondary to the presence of the hilar tumor, so use of other technologies such as spyglass has not been possible.

We initially published a comparison of results between RTX and LTX in 2005.[9] This study compared 38 patients who underwent LTX to 26 patients who underwent RTX, but we included patients with both HC arising in the setting of PSC as well as de novo HC. Since that time, we have entered more than 200 patients in the LTX protocol and over 100 more patients have undergone transplantation. We now recognize that results for patients with HC arising in the setting of PSC are much better than results for patients with de novo HC, both for survival after initiation of neoadjuvant therapy and survival after LTX.[26] These patients have underlying liver disease, multiple strictures, and often undergo endoscopic cholangiography for evaluation of jaundice or treatment of cholangitis. Therefore, we believe our findings suggest that we are more likely to detect HC arising in the setting of PSC earlier in the course of the malignancy. The underlying liver disease and the potential for tumor multicentricity essentially preclude RTX, and these patients are best treated with neoadjuvant therapy and LTX.[27]

An additional focus of this study was to compare RTX with LTX for patients with de novo HC and identify clinicopathologic factors that might useful for operative selection. There were clinicopathologic differences between treatment groups. Patients undergoing RTX were older, had significantly larger and more poorly differentiated HC, and more frequent lymph node metastases (by selection) than patients undergoing LTX. Patients undergoing

RTX were older than those who underwent LTX. This difference is not surprising because older patients are less likely suitable for LTX. Patients in the RTX group had a lower negative margin resection rate (R0, 92%) than the LTX group (100%), which one would expect to be due to the nature of the operations. There was a higher rate of nodal involvement in the RTX group compared with the LTX group (38% vs 2%). This finding is also expected because patients in the LTX group underwent operative staging to exclude lymph node metastases. Patients in the RTX group also had a larger mean tumor diameter and a higher proportion of poorly differentiated tumors than were in the LTX group (63% vs 35%, respectively). These differences are difficult to interpret because neoadjuvant therapy is highly effective and known to cause tumor necrosis. Indeed, only 70% of patients with de novo HC were found to have viable cancer on pathologic examination of their explanted livers.[28] Nevertheless, other than lymph node metastases, none of the other clinicopathologic correlates of survival could be reliably used preoperatively for selection.

There are a multitude of issues that confound the decision of whether a patient with de novo HC should undergo operation with an attempted RTX or treated with neoadjuvant therapy and subsequent LTX. With RTX, the only issues to consider are those that directly affect the patient. Even if the likelihood of achieving an R0 resection is low, it is often in the patient's interest to attempt it, especially if the patient is not a suitable candidate for LTX due to older age, comorbid medical conditions, or psychosocial issues.

In contrast, the threshold for proceeding with LTX is much higher than that for RTX. The patient must be healthy enough to survive neoadjuvant therapy and LTX. One must also consider deceased donor organ availability, anticipated waiting time, the possibility of living donor transplantation and the transplant program's outcomes, which are carefully monitored. There is also uncertainty with the possibility that regional lymph node involvement could be detected at operative staging and preclude LTX. Endoscopic ultrasound-directed aspiration of the regional lymph nodes before initiation of neoadjuvant therapy has decreased the likelihood of finding nodal involvement during the staging operation, but involvement is still encountered 10% to 15% of the time.[26]

Limitations of this study include, first, its retrospective design as well as referral bias to a single center with a multidisciplinary interest in HC. Second, overall treatment differed not only by operation but neoadjuvant therapy. Although adjuvant chemotherapy was used with increasing frequency after resection of node-positive HC during the study, such treatment was not standard and neoadjuvant therapy was never used in the RTX group. Third, comparison of TNM staging, which is a major predictor of survival, was not possible because of the neoadjuvant therapy in the LTX group. Finally, survival after LTX may not reflect outcomes from other centers because our center has the largest experience in LTX for HC.

CONCLUSIONS

We conclude that patients with clearly resectable de novo HC should be treated with RTX because there is no evidence to date that they would fare better with neoadjuvant therapy and LTX. Patients with locally unresectable de novo HC, and meeting the previously mentioned criteria, should be treated with neoadjuvant therapy and LTX. Survival after initiation of neoadjuvant therapy and subsequent LTX is clearly sufficient to justify use of a deceased or living donor liver. The decision to proceed with RTX or LTX for patients with borderline resectable de novo HC remains unclear and difficult. The decision process should consider the center's track record predicting resectability and lymph node involvement, the availability of either a deceased donor or living donor graft, and the individual patient's suitability for aggressive neoadjuvant therapy and LTX. Our results also suggest that some patients with borderline B-C type III HC might be best treated with LTX, especially if they are excellent transplant candidates.

Author Contributions

Study conception and design: Croome, Rosen, Heimbach, Nagorney
Acquisition of data: Croome, Nagorney
Analysis and interpretation of data: Croome, Nagorney
Drafting of manuscript: Croome, Rosen, Heimbach, Nagorney
Critical revision: Croome, Rosen, Heimbach, Nagorney

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Discussion

DR ROBERT CG MARTIN (Louisville, KY): Dr Croome and authors, from the Mayo Clinic Division of Transplantation Surgery, present their retrospective review of 99 patients who underwent resection and 54 patients who underwent liver transplantation for newly diagnosed de novo hilar cholangiocarcinoma. This institution has been an innovative leader in the use of liver transplantation for patients with hilar cholangiocarcinoma, especially in patients with underlying primary sclerosing cholangitis (PSC). They have clearly demonstrated that patients who present with hilar cholangiocarcinoma and PSC do show a significant improvement in overall survival when compared with hepatic resection. This retrospective review is an evaluation of patients with Bismuth-Corlette-based type IV cholangiocarcinoma as well as Bismuth-Corlette type IIIb.

The authors have reported similar overall survival in patients with either resection or transplantation who had no lymph node metastasis and were able to achieve a margin-negative resection. They also demonstrated that patients who, at the time of resection, were reclassified as Bismuth-Corlette type IV did have an inferior survival to patients who underwent liver transplantation with lymph-node-negative, well-differentiated hilar cholangiocarcinoma who were preoperatively classified as Bismuth-Corlette type IV. I have 3 key questions for the authors that appear to relate to this innovative evaluation.

First, given that nearly 40 patients who were preoperatively thought to be Bismuth-Corlette type III were reclassified as Bismuth-Corlette type IV, there appears to be a significant limitation in current preoperative imaging and staging because nearly 50% of resection candidates would have potentially been eligible for liver transplantation simply based on their biliary involvement. Therefore, what have you done or what do you recommend for other hepatobiliary units in regard to enhancing the overall biliary involvement sensitivity and accuracy? Are you more liberally using intrabiliary endoscopy or the spyglass type technique to better evaluate intraluminal secondary and tertiary biliary involvement before decision making on resection or liver transplantation?

Second, you conclude that patients with unresectable de novo hilar cholangiocarcinoma should be treated with neoadjuvant therapy and liver transplantation; however, your results do not confirm that